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Research Statement

Research interests: Applied Microeconomics, Production Econometrics, and Industrial Organization.

I study how technology adoption changes firm productivity and market power, and why its benefits differ across firms. My research combines economic models with firm-level data to distinguish gains in productive efficiency from changes in pricing behavior. My independent agenda develops two connected directions: recovering reliable economic quantities from flexible production models, and identifying how AI-related productivity gains spread beyond adopting firms. These solo projects extend my dissertation toward questions about measurement, complementary investments, and the distribution of technology's benefits. I also have two coauthored publications: a study of productivity in Nigerian banking in *Economic Modelling* and a study of heterogeneous treatment effects in *Economics Letters*.

Job Market Paper: Digitalization, Factor-Biased Productivity, and Market Power

My job market paper (with Subal C. Kumbhakar) distinguishes three channels through which digitalization changes firm performance: labor efficiency, efficiency across the input bundle, and markups. Its sequential identification framework allows persistent productivity and markups, endogenous digitalization, and imperfect competition without imposing a particular model of competition.

The application uses the ESEE panel of Spanish manufacturing firms, 2000–2019, and a thirteen-variable digitalization index covering IT, digital human capital, automation, and interaction with customers and suppliers. Under imperfect competition, a 0.10-point increase in the index is associated with **0.55 percent higher labor-augmenting productivity**, **0.43 percent higher Hicks-neutral productivity**, and **0.07 percent higher markups** in the following period.

Labor-augmenting productivity measures labor efficiency due to technical change. Its response is significantly larger for small firms under both perfect and imperfect competition; the markup response is significantly smaller in highly digitalized industries. These conditional relationships show why evaluating technology adoption requires distinguishing productivity gains from pricing responses.

My Independent Research Agenda

My two solo projects extend this work to the reliability of productivity estimates and the transmission of technology's benefits across firms.

Regularization in Translog Production Models. I ask whether models selected for accurate output prediction also yield reliable input elasticities, returns to scale, and markups. Correlated squared and interaction terms can make these economic quantities unstable even when predictions are accurate.

I am developing a comparison of Lasso, Ridge, Elastic Net, principal component regression, and partial least squares. Planned simulations will vary input correlations, sample size, and production relationships. I will assess elasticity bias, returns-to-scale accuracy, implausible marginal products, and resulting markup errors. I will also examine these methods alongside approaches to endogenous input choices, which regularization alone does not resolve. The aim is to guide method choice when economic interpretation matters.

Artificial Intelligence and Spatial Factor-Biased Productivity. This planned project asks whether AI improves the efficiency of particular inputs and whether those gains extend to other firms through knowledge flows, worker mobility, and supply chains.

I will first define measures of firms' AI applications and the production activities they affect. I will then develop a design to distinguish spillovers from common shocks while accounting for endogenous adoption and input choices. Comparing geographic proximity with industry and supply-chain exposure will help identify who benefits and which complementary investments matter.

Together, these projects connect credible productivity measurement with technology's uneven returns. Longer term, I plan to combine heterogeneous treatment-effect methods with production models to study differences in adoption gains.

Related Research

In **“The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking”** (with Subal C. Kumbhakar; *Economic Modelling*, 2026), we study seventeen banks over 2000–2023. Productivity declined by an estimated 3.5 percent annually despite cost-reducing technical change and returns to scale averaging 1.46. This gap motivates my interest in how firms translate technological possibilities into realized performance.

In **“Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects”** (with Ivan Korolev and Nancy Dhameja; *Economics Letters*, 2026), simulations show why method choice should reflect the structure of effect heterogeneity. This informs my focus on matching empirical tools to economic questions.

In **“Mediation Analysis in Production Functions with Unobserved Productivity”** (with Subal C. Kumbhakar), we distinguish digitalization's relationship with output through labor hours from changes in efficiency. Controlling for unobserved productivity in both equations addresses a source of overstated indirect effects.